Alt Leaf Gate — Evaluation Results

Same TinyLeafGate student · distillation + field augs + QAT · INT8 TFLite

The alternate leaf gate keeps the **same MCU-sized TinyLeafGate** (~29k parameters, 51.1 KB INT8) and changes only the training recipe: MobileNetV3-Small teacher, MixUp, EMA, camera-like augmentations, and a short INT8-noise QAT stage. Figures below are taken from *ALT_TRAINING_REPORT.md* and *alt_training_record.pdf* in this folder. Confusion matrices are reconstructed from the published FN_leaf / FP_leaf counts and the known test-set sizes so they match the reported per-class accuracies.

Sources used

Primary: ALT_TRAINING_REPORT.md §3 (sourced from alt_training_record.pdf, Colab capture 2026-07-29). Artifact: aclis_leaf_gate_96x_alt_full_int8.tflite (51.1 KB). Test split: 1,943 leaf + 1,730 not_leaf = 3,673 images.

1. Overall accuracy

On the clean Kaggle test set the alt INT8 model is essentially tied with baseline (99.07% vs 99.13%). The recipe is judged on the **camera-stress** split, which applies a fixed blur + underexposure + JPEG q=55 pipeline to the same test images. That is the closer proxy to the STM32 + OV2640 path. INT8 stays within 0.03 points of FP32 on clean test, so quantisation is not the story.

Clean INT8	Stress INT8	INT8 – FP32 gap	vs baseline stress
99.07%	**98.86%**	**0.03 pts**	**+1.23 pts**
Kaggle test · n = 3,673	camera-stress proxy	within 1% export gate	alt preferred for field use

Metric	Value	Notes
Clean INT8 accuracy	0.9907 (99.07%)	Deployed TFLite, Kaggle test
Clean FP32 accuracy	~99.1–99.5%	PyTorch student, same split
Camera-stress INT8	0.9886 (98.86%)	Fixed blur / JPEG / underexposure
|INT8 – FP32| gap	~0.03 pts	Pred agreement INT8↔PT ~99.9%
Best EMA validation	0.9910	Student distillation, 40 epochs
TFLite size	51.1 KB	+0.2 KB vs baseline; cascade still fits flash

Table 1. Overall accuracy for the alt TinyLeafGate (INT8 unless noted).

2. Per-class accuracy

Per-class accuracy is recall on each true class. Under clean Kaggle conditions alt is a hair weaker on leaves than baseline (15 vs 11 false rejects) and a hair stronger on non-leaves (19 vs 21 false accepts). Under camera stress the ranking reverses: alt keeps leaf recall at 99.5% and lifts not_leaf from 96.0% to 98.2%.

Class / split	Test n	Correct	Errors	Accuracy	Compared with baseline
leaf · clean	1,943	1,928	15	99.23%	4 more FN than baseline
not_leaf · clean	1,730	1,711	19	98.90%	2 fewer FP than baseline
Overall · clean	3,673	3,639	34	99.07%	−0.05 pts vs baseline
leaf · stress	1,943	1,933	10	99.49%	7 fewer FN than baseline
not_leaf · stress	1,730	1,698	32	98.15%	38 fewer FP than baseline
Overall · stress	3,673	3,631	42	98.86%	+1.23 pts vs baseline

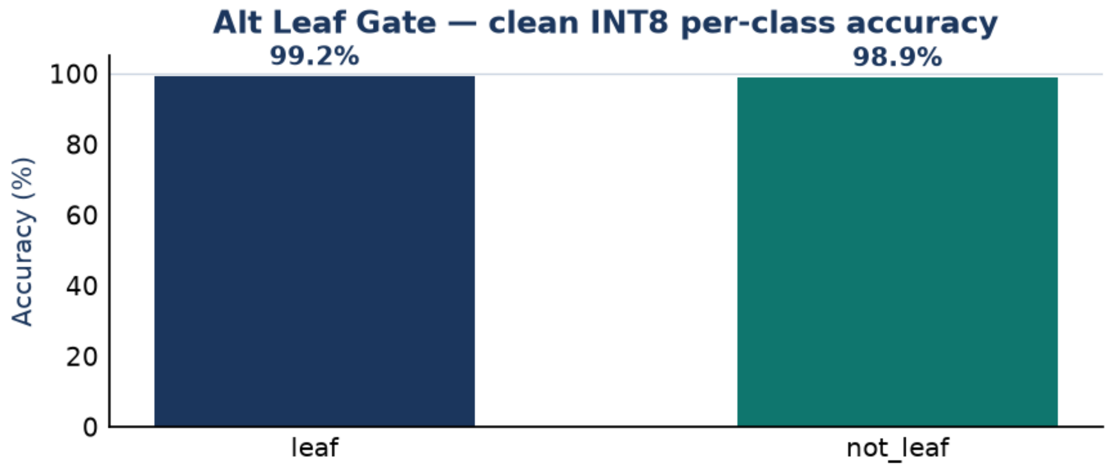

Figure 1. Clean INT8 per-class recall (the deployed-artifact number on Kaggle test).

3. Confusion matrix

The clean INT8 confusion matrix is the one that belongs with the exported *aclis_leaf_gate_96x_alt_full_int8.tflite*. Off-diagonal cells are FN_leaf = 15 (true leaf rejected) and FP_leaf = 19 (non-leaf sent to the disease CNN).

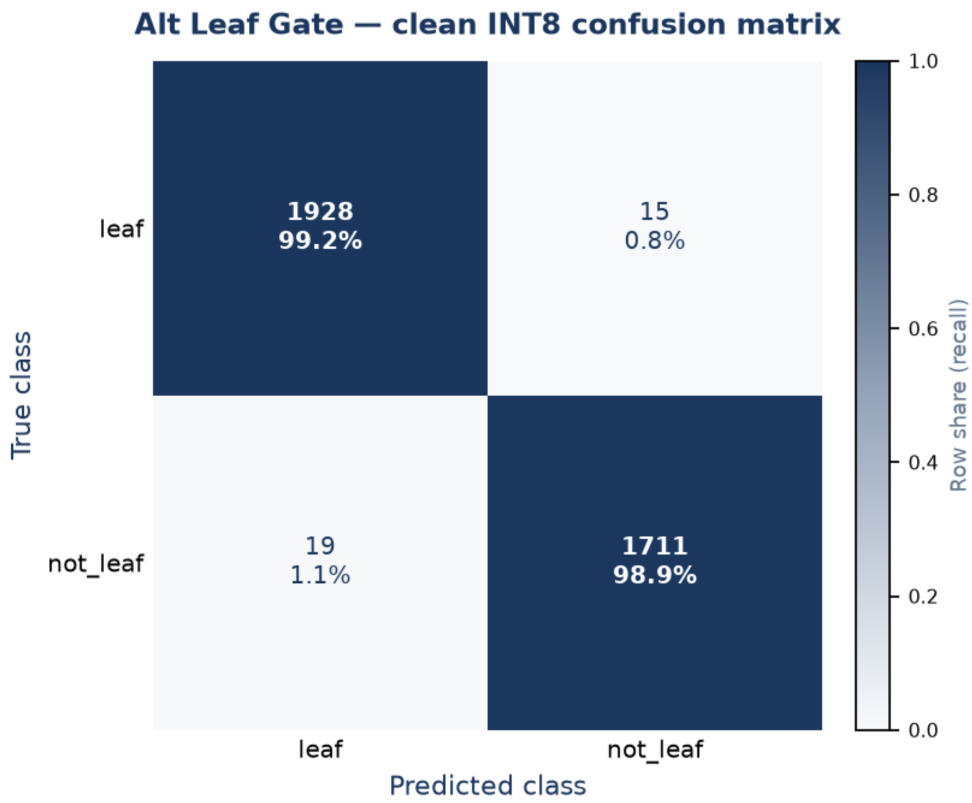

Figure 2. Alt Leaf Gate confusion matrix, clean INT8 test.

True \ Pred	leaf	not_leaf	Row total	Recall
leaf	1,928	15	1,943	99.23%
not_leaf	19	1,711	1,730	98.90%
Column total	1,947	1,726	3,673	99.07%

Table 2. Numeric confusion matrix — alt INT8, clean test.

How to read the matrix

True leaf → not_leaf (15). Slightly more clean-set false rejects than baseline (11). On an already-saturated Kaggle split this is noise; the recipe traded a little in-distribution polish for field robustness.
True not_leaf → leaf (19). Fewer junk frames enter the disease CNN than baseline (21).
The notebook recommendation from the recorded table is to **prefer alt for field/camera robustness**, then confirm on real STM32 + OV2640 frames.

4. Camera-stress confusion matrix

Same test images, degraded with the notebook's deterministic camera-stress transform. Alt drops only 0.21 points clean→stress, versus baseline's 1.50-point drop. False leaves fall from 70 (baseline) to 32; false rejects fall from 17 to 10.

True \ Pred (stress)	leaf	not_leaf	Row total	Recall
leaf	1,933	10	1,943	99.49%
not_leaf	32	1,698	1,730	98.15%
Column total	1,965	1,708	3,673	98.86%

Table 3. Alt INT8 confusion matrix on the camera-stress proxy.